

Logan Burns

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Education

University of Florida, B.S. Electrical Engineering

Expected May 2028

Herbert Wertheim College of Engineering, GPA 3.91

Gainesville, FL

- University Honors Program; University Research Scholars Program; FOCUS Lab REU award. Graduate coursework: Analog IC Design I (EEE 5320).

Research Experience

CHIP Lab, UF ECE

Sep 2026 – Present

Undergraduate Research Assistant (PI: Dr. Hamed Dalir)

Gainesville, FL

- Photonic data converters for a silicon-photonics AI accelerator; synthesized the microring, phase-change-material, and quantization-codesign literature into a design-space reference for the group's DAC and ADC threads, targeting electron-beam SOI fabrication.

Florida Optics and Computational Sensor (FOCUS) Lab, UF ECE

Sep 2025 – Present

Undergraduate Research Assistant (PI: Dr. Sanjeev Koppal)

Gainesville, FL

- Co-authored *EdgeTrain*, a thermal-adaptive continual-learning controller for robot-mounted edge GPUs; built the thermal instrumentation and evidence base behind its central claims.
- Built the calibrated FLIR A6751sc imaging pipeline and 150 W radiant-heat stress rig characterizing continual-learning workloads on a Jetson Orin NX, isolating a 20 °C thermal-signature difference between algorithms; the controller ran 16.3 °C cooler at equal accuracy with zero throttle events.
- Sole author of a virtual thermal sensing system predicting off-die component temperatures (VRMs, MOSFETs) from on-die telemetry alone, needing no camera after a one-time calibration: watershed segmentation, lumped-capacitance fitting (L-BFGS-B), real-time daemon under 1% CPU. Predicts 16 hotspots to 0.34 °C mean absolute error on held-out configurations.
- Extended it from point prediction to the full thermal field: eight POD modes capture 99.8% of the spatial variance, reconstructing every pixel to 0.41 °C mean error on held-out configurations.

Human Systems Engineering Lab, UF: Undergraduate Research Assistant

Jan – May 2025

- Ran experimental trials and built a reproducible R analysis pipeline for a pre/post study on driver-education method and interaction with vehicle automation.

Manuscripts

M. Keivanimehr, **L. Burns**, S. J. Koppal, “EdgeTrain: Thermal-Adaptive Continual Learning for Energy-Constrained Mobile Robots,” manuscript in revision for conference resubmission, 2026.

Teaching

Head Peer Instructor, EEL 3701C Digital Logic & EEL 4744C μ P Applications

Spr 2026 – Present

Leads the peer-instructor team for both courses, UF ECE

Gainesville, FL

- Lead help sessions on external-bus memory mapping, address decoding, and digital logic; author lab specifications, solutions, and course documentation.
- Manage exam-grading assignments across the PI team, hardware inventory, and repair coordination; point of contact for fellow peer instructors.
- Remediated 150 course documents for ADA/WCAG accessibility with a custom Python toolchain; resolved a Quartus–Questa incompatibility with a C++ shim distributed to students.

Design Teams & Fabrication

Machine Intelligence Laboratory (MIL), UF: SubjuGator 9 AUV

Jan 2026 – May 2026

- Sole designer of the leak-detection subsystem: LTspice comparator front end through Altium schematic and layout to a fabrication-released board; assembled and brought up v1 myself by SMT reflow.

Semiconductor Readiness Organization, UF Nanoscale Research Facility

Sep 2024 – Present

- Fabricated MOS capacitors in the NRF cleanroom: dry oxidation, spin-coating, photolithography, HF wet etch, sputter deposition.

Selected Projects

Discrete Analog Design: LDO Regulator and Two-Stage Op-Amp (individual)

- Rebuilt a discrete LDO whose original compensation never let the loop cross unity: a series damping zero below crossover holds phase margin within 0.1° across a 60:1 ESR range, and a P-channel pass device restores regulation at the low input corner (12.6 mV line regulation, 8–16 V). Foldback limiting cuts pass-device dissipation 9.3 W to 2 W.
- Designed a discrete two-stage BJT op-amp: 110 dB open-loop gain, 3.63 MHz GBW, 106° phase margin by nulling the right-half-plane zero emitter degeneration drags toward crossover. Both designs are now under bench characterization.

Skills

Languages: C, C++, Python, MATLAB, R, VHDL, SystemVerilog, AVR assembly

EDA & Simulation: Altium Designer, LTspice, Cadence Virtuoso, Quartus, QuestaSim (SVA, CRV)

Embedded & Lab: NVIDIA Jetson, PyTorch, FLIR/GigE Vision, DMA/interrupt-driven firmware, Linux, Git; thermal imaging, I–V characterization, oscilloscope/AWG/network analysis; SMT reflow; cleanroom microfabrication